

Assessment Project

Team Task Management System

Role-Based Full-Stack Application Assessment

1. Project Brief

Build a role-based Task Management System that allows organizations to manage teams, assign tasks, track task progress, collaborate through comments, and notify users of important task events.

2. User Roles & Capabilities

Role	Capabilities
Admin	Manage teams and assign tasks to managers and users.
Manager	Create and assign tasks to team members.
User	View and manage assigned tasks.

3. Feature Requirements

Authentication & Authorization

- JWT or OAuth-based login.
- Role-based access control for Admin, Manager, and User.
- Secure password hashing.
- Session expiration / token expiration handling.

Core Functionalities

- User registration and login.
- Task creation, assignment, and status tracking.
- Supported task statuses: To Do, In Progress, Done.
- Team management, including assigning users to teams by managers.
- Comment section for task collaboration.

Notifications

Email notifications, or mock notifications if email integration is not implemented, must be triggered by the following events:

- Task assignment.
- Task status update.

Dashboard

- Overview of task statuses per user.
- Filtering by deadline, status, and priority.

4. Technology Stack

Layer	Required / Preferred Technology
Backend	Node.js with Express, or .NET . .NET implementation is considered a bonus.
API Style	REST API with proper authentication, authorization, validation, and error handling.
Database	SQL Server or PostgreSQL.
ORM	Entity Framework / Entity Framework Core for .NET implementations.
Frontend	React with Axios for API integration.
UI	Clean, responsive UI with API integration and appropriate error handling.

5. Bonus / Advanced Expectations

- Full-stack Dockerized setup using a multi-container architecture.
- Unit test cases and/or integration tests.
- Swagger or Postman API documentation.
- CI/CD setup using GitHub Actions.
- Deployment to Render, Railway, Netlify, or Vercel.
- Mobile-responsive and accessible UI.

6. Deliverables

- **GitHub repository** containing well-organized frontend and backend code.
- **README.md** including setup instructions, sample credentials, and the technology stack used.
- **API documentation** using a Postman collection or Swagger documentation.
- **Optional live demo** with a deployed application link.
- **5–8 minute video walkthrough** recorded using Loom, Google Drive, or a similar platform.

7. Deadline

Please complete and share all deliverables as early as possible. If additional time is required, please communicate the request for a small extension.

8. Evaluation Criteria

Section	Points
Backend API Design & Auth	20
Database Design & Relations	15
Frontend UI & UX	15
Role-Based Access & Logic	10
Code Quality & Modularity	10
Testing (Unit / Integration)	10

Section	Points
DevOps (Docker / CI/CD / Deployment)	10
Documentation & Presentation	10
Total	100

Assessment Goal: Demonstrate the ability to design, implement, secure, test, document, and deploy a production-oriented role-based full-stack task management application.